PROJECT

BCE-P663

Submitted By:
Amarpreet kumar (236301033)
Ayush Aryan(236301063)

B.Tech VI Sem (III-Year)

Submitted To:
Dr Nishant
Munjal sir

Assist. Professor
CSE Department, FET, GKV

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

FACULTY OF ENGINEERING AND TECHNOLOGY GURUKULA KANGRI (DEEMED TO BE UNIVERSITY)

HARIDWAR (2025-2026)

CERTIFICATE BY STUDENT

This is to certify that the project titled **"HostelCare – A Hostel Complaint Management System"** submitted by **Amarpreet Kumar (236301033)** and **Ayush Aryan (236301063)** of B.Tech Computer Science, Third Year, Faculty of Engineering and Technology, Gurukula Kangri (Deemed to be University), Haridwar, is a bonafide work carried out under my supervision during the academic session 2025–2026.

Mentor Signature: _____

Dr. Nishant Munjal
Department of Computer Science
Faculty of Engineering and Technology
Gurukula Kangri University

ABSTRACT

HostelCare is a full-stack web-based Hostel Complaint Management System developed to digitize and simplify hostel issue reporting and maintenance workflows. Traditional complaint systems in hostels are often manual, unorganized, and lack transparency, leading to delayed issue resolution.

This project provides a centralized digital platform where students can register complaints, upload issue-related images, and track complaint progress. Admin users can monitor all complaints, assign them to staff members, and manage workflow efficiently. Staff users can view assigned complaints and update their status after resolution.

The system is developed using the MERN Stack (MongoDB, Express.js, React.js, Node.js) with JWT Authentication for security, Multer for image uploads, Tailwind CSS for UI design, and Recharts for analytics dashboards.

HostelCare improves operational efficiency, accountability, transparency, and communication between students, hostel administration, and maintenance staff.

1. INTRODUCTION

HostelCare is designed as a digital complaint lifecycle management platform specifically for hostel environments. It aims to streamline the process of complaint registration, monitoring, assignment, and resolution.

The system eliminates traditional paper-based or verbal complaint methods and introduces a role-based secure platform with:

- Student Complaint Registration
- Admin Monitoring
- Staff Resolution Workflow
- Complaint Analytics Dashboard

2. PROBLEM STATEMENT

In traditional hostel systems:

- Complaints are manually registered
- No proper tracking system exists
- Delayed complaint resolution
- Lack of accountability
- Poor communication between stakeholders

This causes inefficiency and dissatisfaction among hostel residents.

3. EXISTING SYSTEM

Existing Manual Process:

- Students verbally report issues ,Complaint may be forgotten
- No digital records , No real-time tracking , Resolution delays
- Drawbacks:
- Time-consuming and Unstructured

4. PROPOSED SYSTEM

HostelCare introduces:

- Online complaint registration
- Secure authentication
- Role-based access (Student/Admin/Staff)
- Image upload support
- Complaint assignment system
- Status tracking
- Dashboard analytics

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5. OBJECTIVES

- Digitize hostel complaint system
- Improve complaint resolution speed
- Increase transparency
- Maintain complaint records

6. SCOPE OF PROJECT

This system can be used in:

- College hostels
- University campuses
- Residential institutions
- PG accommodations

7. TECHNOLOGY STACK

Frontend:	Backend:	Deployment:
• React.js • Tailwind CSS • React Router DOM • Axios • Recharts	• Node.js • Express.js • MongoDB • Mongoose • JWT Authenticatio	• Netlify (Frontend) • Render (Backend) • MongoDB Atlas (Database)

8. SYSTEM ARCHITECTURE

Student → Complaint Submission → MongoDB Database

Admin → View & Assign Complaint → Staff

Staff → Resolve Complaint → Status Update

Student → Track Updated Status

9. MODULES DESCRIPTION

A. STUDENT MODULE

- Register/Login
- Create Complaint
- Upload Image
- View Own Complaints
- Track Status

B. ADMIN MODULE

- View All Complaints

- Dashboard Analytics
- Assign Complaints to Staff

C. STAFF MODULE

- View Assigned Complaints
- Update Complaint Status
- Mark Resolved

10. DATABASE DESIGN

User Collection :

name

email

password

role

Complaint Collection :

title

description

category

image

status

createdBy

assignedTo

createdAt

updatedAt

11. FUNCTIONAL WORKFLOW

Complaint Lifecycle:

Student creates complaint → Status: Pending

Admin assigns complaint → Status: In Progress

Staff resolves complaint → Status: Resolved

12. IMPLEMENTATION DETAILS

Key Features Implemented:

- JWT Authentication

- Protected Routes
- Role-Based Access Control
- Complaint CRUD
- Image Upload with Multer
- Axios Interceptors
- Dashboard Pie Charts
- Responsive Design

13. TESTING AND RESULTS

Tested Features:

- Registration System
- Login Authentication
- Complaint Creation
- Image Upload
- Admin Assignment
- Staff Resolution
- Dashboard Accuracy

Result:

All major functionalities were successfully implemented and tested.

14. ADVANTAGES

- Paperless complaint system
- Faster issue resolution
- Centralized data
- Better transparency
- Easy complaint tracking
- Improved communication

15. LIMITATIONS

- Requires internet connectivity
- Local image storage limitations
- No push notifications
- Manual staff assignment

16. FUTURE SCOPE

- Cloudinary integration

- Mobile app development
- AI-based complaint categorization
- Email/SMS alerts
- Real-time notifications
- Priority tagging

17. CONCLUSION

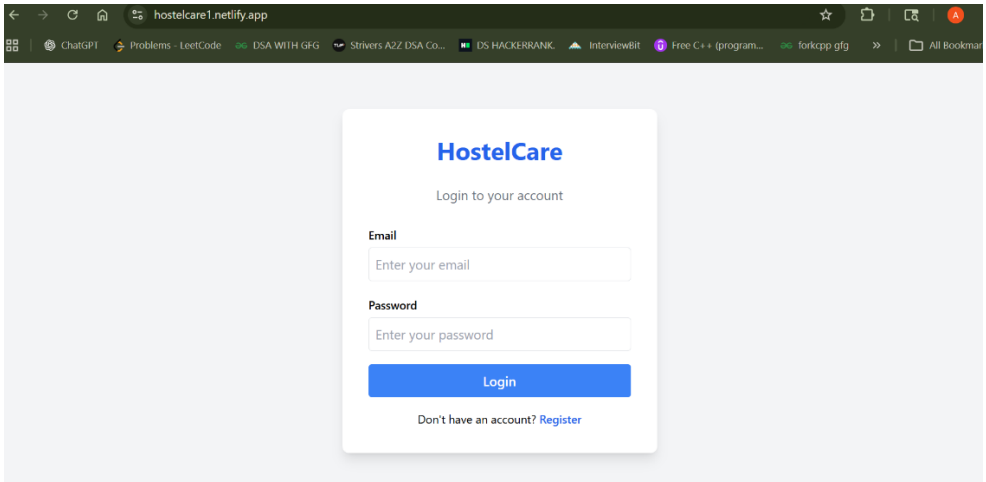
HostelCare successfully addresses the inefficiencies of traditional hostel complaint systems by introducing a structured, transparent, and role-based digital complaint management solution.

By integrating MERN stack technologies, authentication, image uploads, and dashboard analytics, the project demonstrates practical implementation of full-stack development principles while solving a real-world institutional problem.

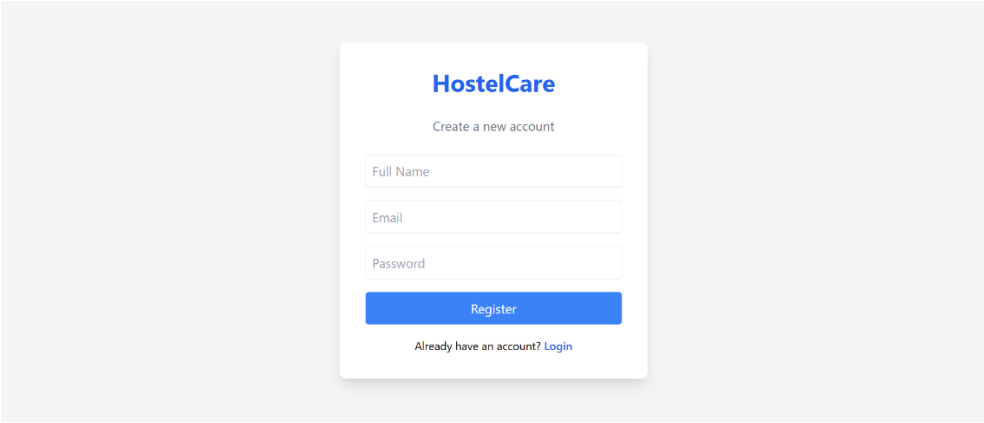
This project enhances technical understanding as well as practical deployment experience.

18.ScreenShots :

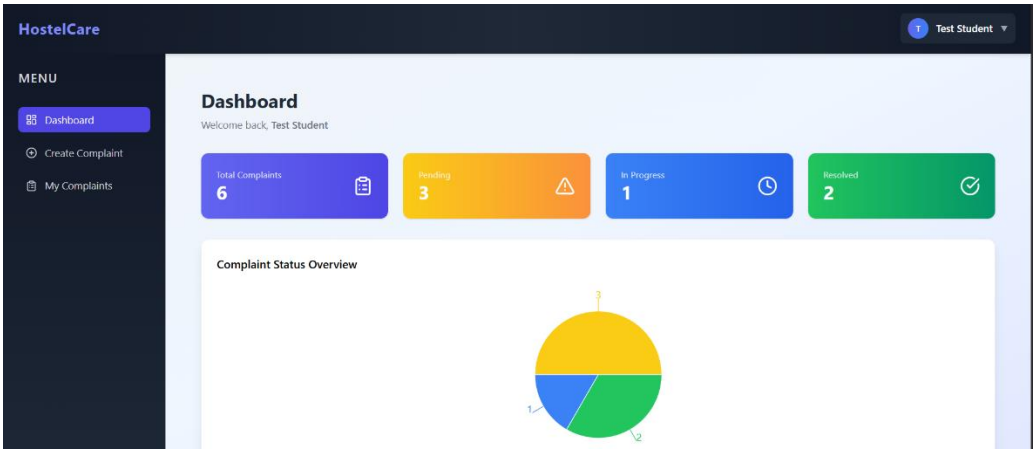
Login Page :



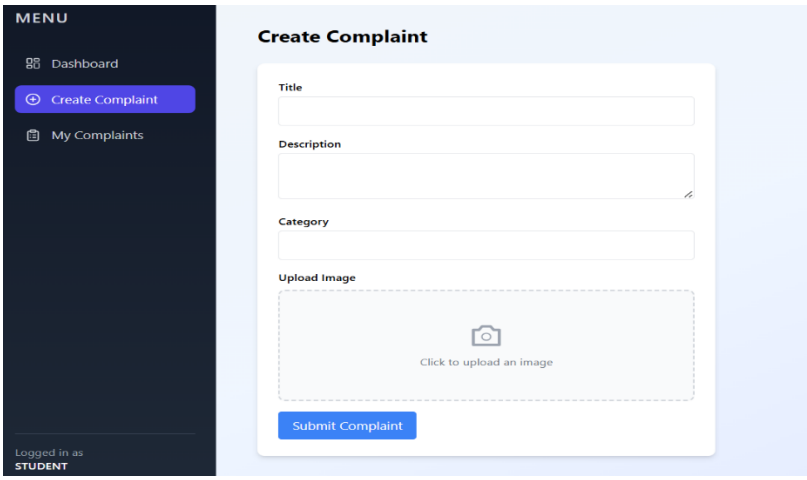
Registration Page :



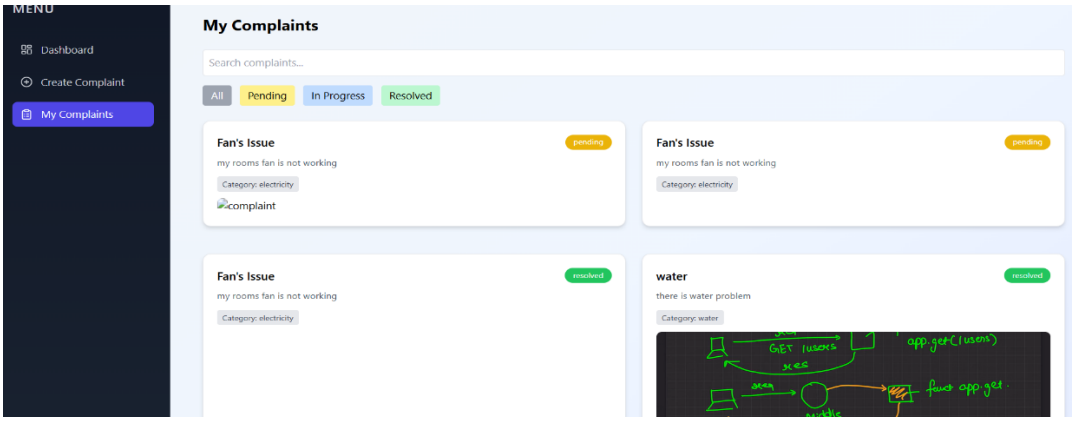
Student Dashboard



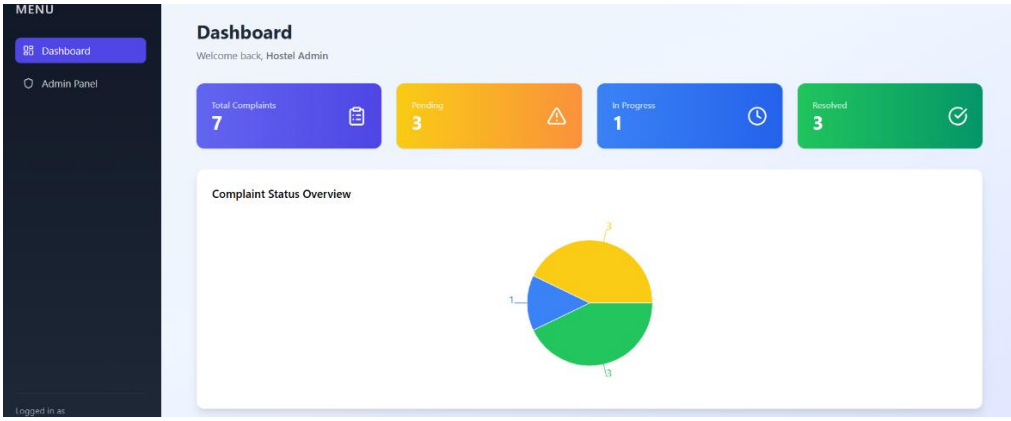
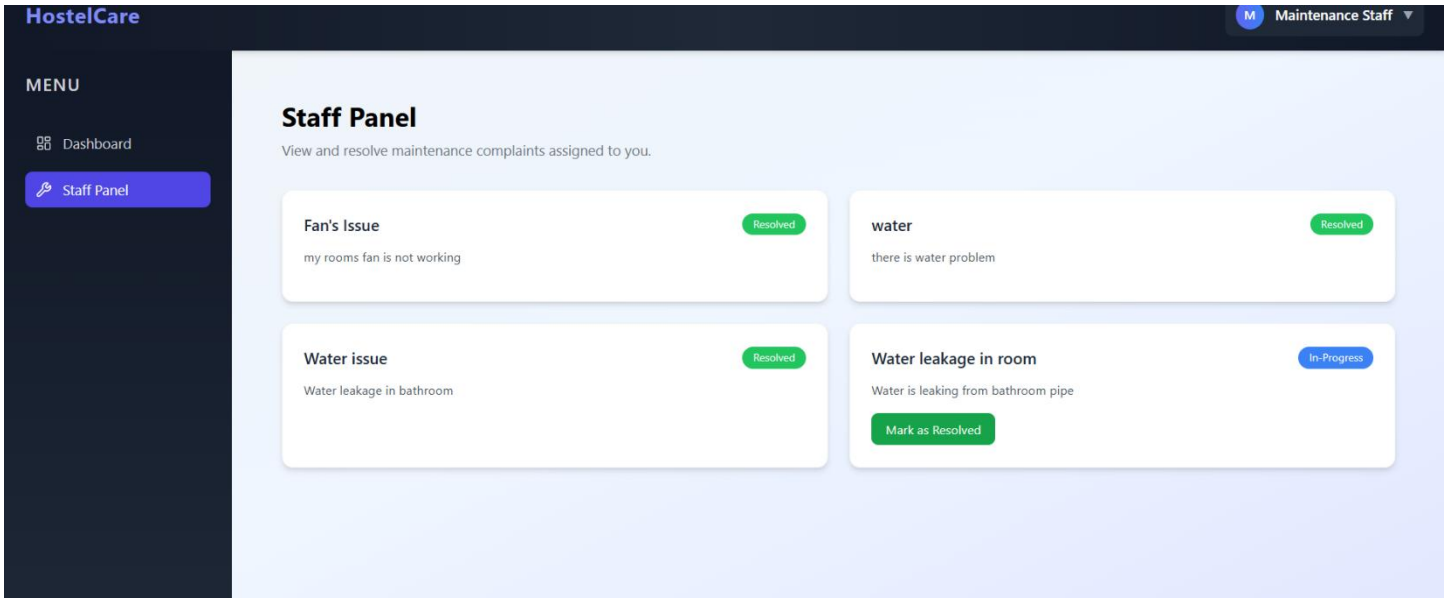
Complaint Creation Form



My Complaints Page



Admin Dashboard



Staff Dashboard :